AWARe

Security-Aware Minimal Neural Adaptation for Longitudinal Keystroke Authentication

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Section A - Review of Three Research Papers

These papers were selected to establish three points needed by AWARe: behavioral biometrics do change over time, adaptive keystroke authentication already exists, and long-term deployment requires adaptation without sacrificing security. The project's contribution is the decision policy that minimizes model change while enforcing recovery, retention, and security constraints.

1. Adaptability of Current Keystroke and Mouse Behavioral Biometric Systems: A Survey Computers & Security, 2026 [1]

Subash et al. review studies on keystroke and mouse behavioral biometrics, focusing on whether existing systems adapt as user behavior evolves. The survey identifies long-term reliability, adaptability, poisoning resistance, and scalable continual learning as open problems.

Strengths. It is a recent survey focused directly on the project's central issue—behavior evolution—and explicitly connects adaptability with security and long-term deployment.

Weaknesses. As a survey, it does not provide a new adaptive algorithm or a controlled head-to-head experiment showing which neural update strategy best preserves security.

2. Double Serial Adaptation Mechanism for Keystroke Dynamics Authentication Based on a Single Password Computers & Security, 2019 [2]

Mhenni et al. propose a keystroke authentication strategy that updates both the user template and the decision/adaptation thresholds during operation. The work shows that template adaptation is already an established response to behavioral variation.

Strengths. It directly studies adaptive keystroke authentication and demonstrates that both user references and decision thresholds can be updated rather than kept static.

Weaknesses. The method is template-centered rather than neural-model-centered, and it does not formulate adaptation as minimizing parameter change subject to explicit recovery, retention, and FAR constraints.

3. On the Long-Term Effects of Continuous Keystroke Authentication: Keeping User Frustration Low through Behavior Adaptation. Proceedings of the ACM on IMWUT, 2023 [3]

Huh et al. study long-term continuous keystroke authentication on smartphones. Their real-world evaluation reports increasing false rejections over two months and uses periodic model retraining to keep FRR low while maintaining attack detection.

Strengths. It demonstrates with deployed, longitudinal data that static authentication performance can deteriorate over time and that adaptation is operationally useful.

Weaknesses. Its adaptation relies on periodic retraining rather than automatically selecting the minimum model update required for a particular drift event, and it does not stress-test adaptation with contaminated update streams.

Section B - Project Idea, Research Gap, and Questions

1. Problem Statement

Keystroke biometrics identifies a person from how they type, such as how long keys are held and how much time passes between key events. Unlike a fingerprint, typing behavior can change naturally because of practice, fatigue, injury, keyboard changes, task context, or simply the passage of time. A model trained on early behavior may therefore reject the legitimate user months later even though no attack has occurred.

AWARe will investigate how to adapt a neural keystroke authenticator without automatically retraining the whole model. When later behavior indicates sustained longitudinal drift, AWARe will search for the smallest model update that restores the legitimate user while preserving earlier behavior and keeping impostor acceptance under control. If a small edit is insufficient, the system will progressively broaden the update scope.

2. Research Gap and Relationship to Prior Work

Prior work already covers continual retraining, template updating, threshold adaptation, and long-term behavior adaptation. The proposed research gap is a security-aware control policy that automatically chooses and escalates the minimum neural update necessary under explicit constraints.

3. Nature of the Project

Build robust adaptive keystroke authentication under longitudinal behavioral distribution shift, and test whether a minimal-update controller can recover legitimate users with less model disturbance and less security degradation than unrestricted fine-tuning or retraining.

4. Central Objective

AWARe will seek the smallest parameter update $\Delta\theta$ that achieves a target recovery level while keeping the increase in FAR below a security tolerance and keeping loss of performance on earlier genuine behavior below a retention tolerance. This formulation makes recovery, retention, and security simultaneous requirements rather than separate post-hoc observations.

5. Research Questions

- Temporal degradation: How does verification performance change as the time gap from enrollment increases under natural free-text behavior?
- Adaptive recovery: Can AWARe recover later legitimate behavior using limited recent samples while modifying fewer parameters than unrestricted fine-tuning?

- Retention and security: Can AWARe improve the drifting user without materially degrading earlier genuine behavior or increasing impostor acceptance?
- Adversarial adaptation: How robust is AWARe when the candidate adaptation buffer contains increasing levels of impostor contamination?

6. Proposed Contributions

- A hierarchical minimal-update policy that automatically escalates from localized editing to broader adaptation only when recovery, retention, or security constraints are not satisfied.
- A chronological security evaluation using natural long-term Clarkson II behavior plus controlled KBOC multisession verification, including contaminated adaptation streams.

7. System Components

Module	Input -> Output	Role
Feature pipeline	Timestamped key events → timing vectors	Compute hold time, press-to-press, release-to-press, typing speed, and timing variability; preserve timestamps for chronological splits.
Baseline authenticator	Timing vector/sequence → genuine score	Use a compact feed-forward neural network as the primary model; include one GRU/LSTM only as a secondary sequence baseline.
Drift detector	Recent scores/features → drift signal	Trigger adaptation only after sustained confidence/loss deterioration and/or validated feature-distribution shift.
Confidence gate	Candidate sample → trusted buffer	Admit only high-confidence samples to reduce self-poisoning risk; maintain a bounded recent-sample buffer.
Minimal-update controller	Drift event → update scope	Start with the smallest edit, test constraints, and broaden the update only if recovery is inadequate or constraints are violated.
Security evaluator	Scores + labels → EER/FAR/FRR	Evaluate recovery, retention, impostor acceptance, model displacement, and cost after each candidate update.

8. Attribution-Guided Localized Low-Rank Editing

AWARe will first investigate whether attribution or causal-tracing analysis can identify hidden representations or layers that are disproportionately involved in a drifting user's verification decision. Rather than assuming that drift is stored in a single location, this step tests whether the relevant representation is sufficiently localized to support a targeted intervention.

If AWARe can identify a candidate region, it will experimentally evaluate a low-rank weight update, inspired by ROME's minimal-intervention principle [6], as the first adaptation stage. Standard ROME was developed for localized factual editing in language models and is not

assumed to transfer directly to keystroke biometrics. The purpose is to test whether a small low-rank change at an attribution-identified region can recover the drifting user while minimizing effects on retained genuine behavior and impostor scores.

If attribution does not reveal a stable localized region, or if the low-rank update fails the recovery, retention, or security constraints, the controller will proceed to broader gradient-based adaptation stages. A negative result would still be informative by indicating that user-specific drift is distributed rather than cleanly localized in the shared verifier.

9. Hierarchical Adaptation Policy

Stage 1	Attribution/causal-tracing analysis followed, when localization is sufficiently clear, by a low-rank edit at the identified candidate region.
Stage 2	If constraints fail, update the output layer or final hidden + output layers with a low learning rate.
Stage 3	If still insufficient, broaden to selected-layer incremental fine-tuning.
Stage 4	Use replay-assisted fine-tuning to reduce catastrophic forgetting when broader adaptation is required.
Stop rule	Stop at the first stage that reaches target recovery while retention loss ≤ δ.

10. Datasets

Clarkson II will be the primary dataset because it is aligned with the paper's main claim: adaptation under real long-term longitudinal behavioral change. Published Clarkson II work describes 103 participants, about 12.9 million keystrokes, and passive data collection in an uncontrolled natural environment over roughly 2.5 years. The cleaned release contains 101 user files after personally identifiable information cleanup. The project will preserve chronology rather than randomly mixing early and late behavior.

KBOC will be the secondary controlled benchmark. It contains 300 users across four sessions distributed over about four months, with genuine and impostor attempts. Its repeated fixed-text setting provides a cleaner benchmark for multisession verification and security comparisons, complementing Clarkson II's natural free-text variability.

11. Chronological Clarkson II Protocol

Before fixing exact time windows, a longitudinal coverage report will be generated for every user because Clarkson users have unequal amounts and spans of data. Eligibility criteria will require sufficient samples across multiple time periods. Eligible users will then be split chronologically into Early → Mid-1 → Mid-2 → Late windows, where Early is used for enrollment, later windows are used to quantify temporal degradation and adaptation, and the final window remains held out for post-adaptation evaluation.

The project will use the term longitudinal behavioral distribution shift rather than claiming pure behavioral aging, because changes in Clarkson II may reflect time, task, application, keyboard/context, and natural human variation.

12. Evaluation Metrics

- Biometric performance: EER, FAR, FRR, ROC-AUC, genuine/impostor score separation.
- Adaptation quality: target-user recovery, detection delay, number of recent samples required, and held-out late-window performance.
- Locality / retention: performance change on earlier genuine behavior and, where applicable, unaffected users.
- Model cost: percentage of parameters changed, update time, weight displacement, memory overhead, and number of escalation stages used.
- Poisoning robustness: FAR/FRR/EER and model displacement as impostor contamination increases.

13. Tentative Timeline

Weeks	Phase	Key steps / deliverable
1–2	Data + literature	Obtain/clean Clarkson II and KBOC; build user-level longitudinal coverage report; reproduce static baselines.
3–4	Chronological baseline	Create early/mid/late splits; quantify temporal degradation; finalize drift trigger and confidence gate.
5–6	Localized editing	Investigate attribution/causal tracing and test localized low-rank editing; define recovery/retention/FAR constraints.
7–8	AWARe controller	Implement hierarchical escalation and stopping rule; compare with output-layer and broader fine-tuning.
9–10	Security evaluation	Run replay ablations and 0–30% impostor-contamination experiments on adaptation buffers.
11	Final experiments	Run KBOC replication, sensitivity analysis, figures, tables, and statistical comparisons.
12	Paper preparation	Write results/discussion, limitations, related work, and prepare a submission-ready manuscript.

14. Risks and Mitigation

- Clarkson access / coverage: make Clarkson primary if access is available; retain KBOC as an independent secondary benchmark and implementation fallback.
- Localization / low-rank transfer may not work: attribution may show that drift is distributed across the network, or a low-rank edit may be insufficient. Treat localized editing as the smallest candidate intervention rather than a guaranteed mechanism; if constraints fail, the controller escalates to gradient-based restricted updates.

- False adaptation / poisoning: use confidence gating, bounded buffers, delayed updates, and explicit contaminated-stream stress tests.

15. Expected Outcome

The strongest outcome would show that AWARe recovers longitudinally drifting users while changing fewer parameters and preserving FAR and earlier-session performance better than unrestricted fine-tuning.

References

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